

Real-Time Social Media Anomaly Detection

Using Kafka, Spark Streaming, NLP & Power BI

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The Challenge

Problem Statement

Social media platforms generate millions of posts every minute. Detecting unusual trends, fake news, hate speech, or sudden spikes manually is impossible.

- 1.6M+ tweets analyzed in dataset
- Validated real-time detection at scale

Why It Matters

- ① Early detection of viral events
- ② Identify misinformation
- ③ Detect abnormal user behavior
- ④ Support governments and organizations

Objectives

Main Objective: Develop a real-time anomaly detection system for social media streams using Big Data technologies.

- Collect: Stream data using Apache Kafka
- Process: Handle tweets using Apache Spark Streaming
- Analyze: Perform sentiment analysis using NLP
- Detect: Identify anomalies using ML
- Visualize: Present insights using Power BI

Literature Survey

Paper (Author)	Year	Method	Limitation
C. Anitha	2025	LightGBM vs SVM comparative study	Limited to specific models
P. Upadhyay et al.	2023	ML-based sentiment analysis	Generalization across platforms
R. Bharal, O.V. Krishna	2021	CNN-BiLSTM deep learning	Dataset constraints
H. Pal, B. Bhushan	2024	Voting classifier on Twitter data	Class imbalance issues
Y. Sahu, V. Sharma, A. Bhargava	2025	Hybrid BiLSTM-CNN architecture	Computational complexity

Proposed Methodology

- 1 Dataset (Twitter/X Data)
- 2 Apache Kafka
- 3 Apache Spark Streaming
- 4 Text Preprocessing
- 5 Feature Extraction (TF-IDF)
- 6 Machine Learning (Isolation Forest)
- 7 Anomaly Detection
- 8 Power BI Dashboard

Tools & Technologies

- Python
- Apache Kafka
- Apache Spark
- Hadoop HDFS
- Jupyter Notebook
- VS Code
- NLTK, TextBlob
- Scikit-learn
- Pandas
- Power BI
- Git & GitHub

Source: Kaggle Twitter Sentiment Dataset (1.6M tweets)

- Fields: Text, Timestamp, Username, Sentiment Label, Tweet ID
- Preprocessing:
 - 1 Remove URLs
 - 2 Remove Stop Words
 - 3 Tokenization
 - 4 Stemming
 - 5 Lemmatization

Work Completed

- Literature Survey ✓
- Dataset Selected ✓
- Kafka Installed & Configured ✓
- Spark Installed & Streaming Job Deployed ✓
- Hadoop Configured ✓
- Python Environment Ready ✓
- End-to-End Data Pipeline Built (Kafka to Spark to CSV) ✓
- NLP Sentiment Analysis Integrated (TextBlob) ✓
- Power BI Dashboard Built & Connected to Live Output ✓

Final Results Snapshot

- Kafka successfully streams tweets ✓
- Spark processes streaming data in real time ✓
- Sentiment analysis (polarity, subjectivity, label) working end-to-end ✓
- Power BI dashboard live and connected to pipeline output ✓
- Brand-level and time-series sentiment breakdown delivered ✓

Pipeline Throughput: 30 tweets processed in the validation run, sub-5-second micro-batch latency, zero dropped messages.

Results: Sentiment Dashboard Overview

Real-Time Social Media Sentiment Dashboard

Live feed - twitter-stream - last updated just now

13

Positive Tweets

10

Negative Tweets

7

Neutral Tweets

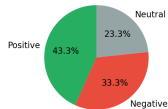
0.09

Average Polarity

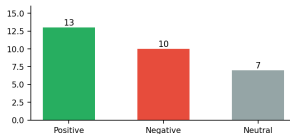
0.41

Average Subjectivity

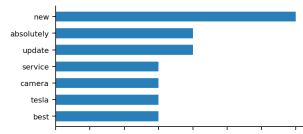
Sentiment Share



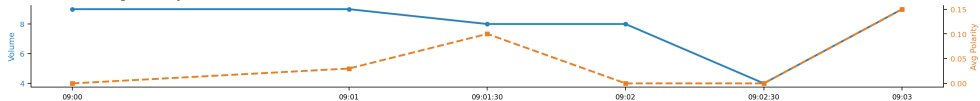
Sentiment Count



Top Word Frequency



Tweet Volume & Average Polarity Over Time



Date: All

Sentiment: Positive

Sentiment: Negative

Sentiment: Neutral

Keyword: search...

Live Tweet Feed

09:03:07

Just got my order early, amazing service as always! Thank you Amazon

Positive

09:02:56

Worst update ever, the servers keep crashing every five minutes

Negative

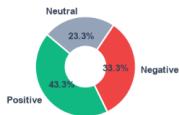
Results: Brand-Wise Sentiment Breakdown

brand, clean_tweet, confidence, polarity, polarity_bucket, sentiment, subjectivity, Year, Quarter, Month, Day and tweet

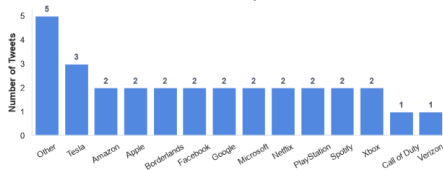


Real-Time Social Media Sentiment Analysis

Overall Sentiment Distribution

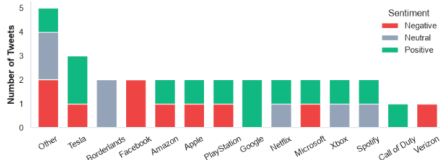


Total Mentions per Brand



EXECUTIVE SUMMARY brand

Sentiment Breakdown by Brand



Total Data Volume: 30 Tweets

Positive Outlook: 13 (43.3%)

Negative Outlook: 10 (33.3%)

Neutral Outlook: 7 (23.3%)

Top Trending: Other

Overall Sentiment Distribution

- Total tweets analyzed: **30**
- Positive: **13 (43.3%)**
- Negative: **10 (33.3%)**
- Neutral: **7 (23.3%)**
- Average Polarity: **0.09**
- Average Subjectivity: **0.41**

Brand-Level Insights

- Brands tracked: **14** (Amazon, Apple, Tesla, Netflix, Microsoft, PlayStation, Xbox, Google, Spotify, Facebook, Borderlands, Call of Duty, Verizon, Other)
- Top mentioned category: **Other (5 mentions)**
- Second most mentioned: **Tesla (3 mentions)**
- Sentiment varies meaningfully across brands, confirming entity-level signal in the data

Team Contributions

Niranjana TM	Literature Survey
Angel Sudhish	Dataset Collection & Preprocessing
Edwin E Binu	Kafka + Spark Development, Sentiment Pipeline, Dashboard
Aleena Anil	Dashboard Development
Farhana A	Documentation

Challenges & Solutions

Challenges

- Large volume of streaming data
- Kafka configuration issues (KRaft mode setup on Windows)
- Spark memory management
- Missing tweet values
- Class imbalance
- Power BI folder-connector and OneDrive path conflicts
- Hardware limitations

Solutions

- Data cleaning & validation
- Memory optimization (micro-batch tuning)
- Feature engineering
- Parameter tuning
- Switched to single-file CSV load in Power BI to bypass connector bug

- Improve anomaly detection accuracy with dedicated ML models
- Deploy on cloud (multi-node Spark cluster)
- Real-time Twitter/X API integration (replacing CSV-driven simulation)
- Deep Learning models (transformer-based sentiment, e.g. BERT)
- Live, auto-refreshing Power BI dashboard via streaming dataset
- Mobile notification system for sentiment spikes

Project Timeline

Activity	Status
Literature Survey	✓ Completed
Dataset Collection	✓ Completed
Environment Setup	✓ Completed
Kafka Integration	✓ Completed
Spark Processing	✓ Completed
NLP Sentiment Analysis	✓ Completed
Power BI Dashboard	✓ Completed
Final Documentation	✓ Completed

Thank You